

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 44

CLIMATE CHANGE - 5

CLIMATE CHANGE

SUSTAINABLE AGRICULTURE

- Income increases for farm households can be made sustainable only if the agricultural practices that do not exhaust/degrade the resources are used. Otherwise, income increases are bound to be short-lived.
- Degradation of natural resources has direct effect on productivity and also on the ability of the farm to withstand biotic and abiotic stresses.
- Loss of Soil Organic Carbon (SOC) and fertility, soil erosion, decline in soil biodiversity, desertification, pesticide pollution, emerging pest resistance and climate change are the major consequences of degradation of natural resources.
- Presently Indian agriculture is facing the critical challenge of feeding a huge growing population under increasingly declining soil quality and changing climatic conditions.



CLIMATE CHANGE

SUSTAINABLE AGRICULTURE

■ Sustainable Agriculture – Definition

- ✓ It is the successful management of resources needed for agriculture to satisfy the human needs, while maintaining or enhancing the quality of the environment, improving the social and economic condition of the farmers and local communities and conserving natural resources.
- **Soil Organic Carbon (SOC)** - has a strong influence on the physical, chemical and biological properties of soils. SOC is a very important factor for soil productivity.

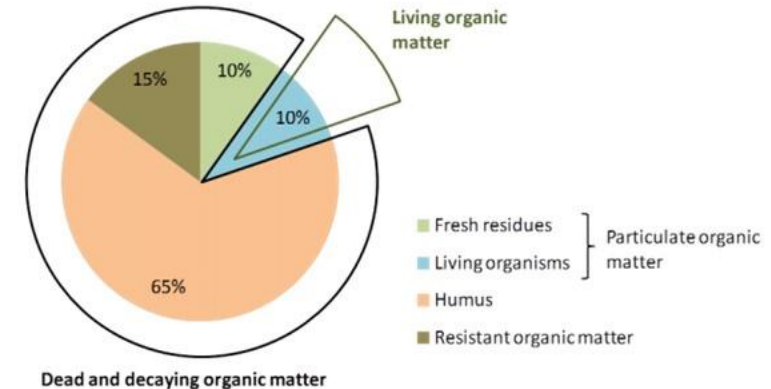


CLIMATE CHANGE

WHAT IS SOC?



- A large amount of carbon is stored in soils, mostly as part of **Soil Organic Matter (SOM)**. SOM makes up around 2-10% of soil's mass and has an important role in the physical, chemical and biological properties of soil.
- SOM **contributes to soil structure, moisture retention, nutrient retention, degradation of pollutants (bio-remediation), carbon sequestration and soil resilience.**
- SOC refers to the carbon component of SOM.
- SOM is **composed of mainly carbon, hydrogen and oxygen and also small quantities of other elements such as nitrogen, phosphorous, sulfur, potassium, calcium and magnesium.**
- SOM is composed of both living and dead organic matter. SOM is a mix of decomposing plant and animal matter, microbes, nematodes (earthworm) etc.



CLIMATE CHANGE

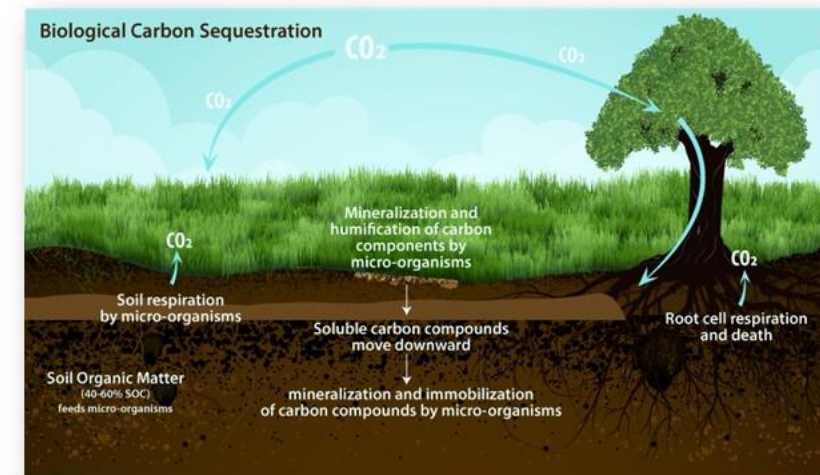
SOC AND CARBON SEQUESTRATION

- Large amount of carbon is stored in soils mostly as SOC. The amount of carbon stored in soils worldwide is approximately three times greater than the carbon presently in atmosphere.
- One of the recommended ways to reduce atmospheric CO₂ concentrations is to increase the global storage of carbon in soils. An added benefit to this solution is the potential for simultaneous enhancement in agricultural production.



BENEFITS OF SOIL SEQUESTRATION OF CARBON

- Fighting climate change, improved soil fertility, reduced soil erosion, increased water retention and greater crop production.
- Some agricultural practices that can enhance carbon storage are - Conservation tillage, cover cropping and crop rotation.
- These are some of the sustainable agriculture practices.

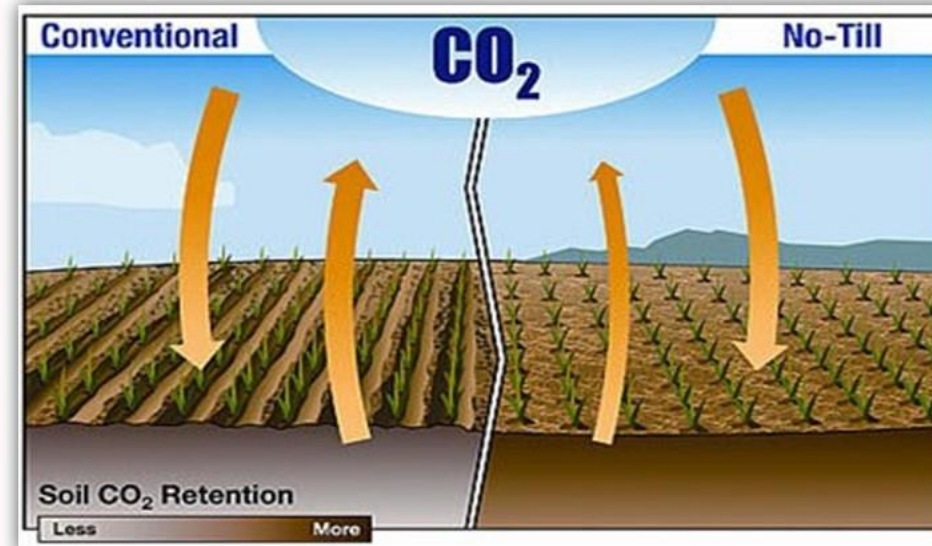


CLIMATE CHANGE

MEASURES FOR MANAGEMENT OF SOIL ORGANIC CARBON (SOC)

1

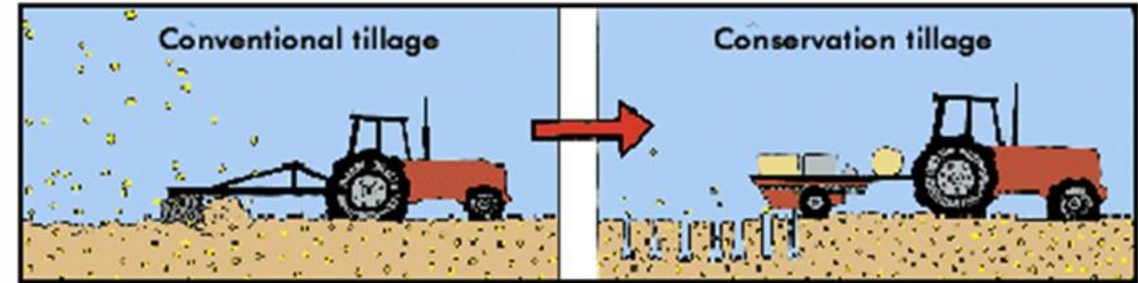
- Maintenance of soil structure in any soil type strongly influences SOC residence times so disturbance (tillage) can lead to substantial losses of SOC.
 - ✓ Frequent disturbance to the soil exposes protected organic matter and increases the rate of decomposition.
 - ✓ So, Carbon sequestration can occur through a reduction in soil disturbance because more carbon is lost from tilled soils than from soils that are less disturbed.
 - ✓ Measures for reducing soil disturbance - Zero Tillage and Conservation Tillage cropping practices.



MEASURES FOR MANAGEMENT OF SOIL ORGANIC CARBON (SOC)

CONSERVATION TILLAGE

- Seedbed preparation, through mechanical soil tillage, is a principal factor responsible for loss of SOC through decomposition.
- Several experiments have shown that ploughing decreases SOC content.
- Conservation tillage practices reduce frequency and intensity of tillage, retain crop residues as mulch on the soil surface, reduce risks of runoff and soil erosion.
- Conservation tillage has a positive impact on activity and species diversity of soil organisms. Also earthworm activity is notably improved by conservation tillage.



MEASURES FOR MANAGEMENT OF SOIL ORGANIC CARBON (SOC)

2 COVER CROPS AND FALLOWING

- A cover crop is a crop of a specific plant that is grown primarily for the benefit of the soil.
- Cover crops are commonly used to suppress weeds, manage soil erosion, help build and improve soil fertility and quality, and control diseases and pests.
- Cover crops are typically grasses. Cover crops add organic matter and nitrogen to the soil.



MEASURES FOR MANAGEMENT OF SOIL ORGANIC CARBON (SOC)

3 RESIDUE MANAGEMENT & MULCHING

- Fields that produce a large quantity of biomass and return it to the soil contain more SOC.
- Improving the humus content is an important strategy to enhance the SOC.
- Practices to enhance humification includes non burning of agricultural residue and returning of crop residue to soil.
- Crop residues are good sources of plant nutrients. It is an excellent source of organic matter as well as rich in NPK.



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299